Abstract

1 Introduction

The provided benchmarks are weighted model counting encoding of quantum circuits from MQTBench benchmark set[2]. By solving the weighted model counting problem, one can simulate a universal quantum circuit with measurement on the first qubit and get the probability of measuring outcome being 0. The encoding method is given in the paper[1].

2 Generator

Users can generate the benchmarks by installing a Python library called `Quokka-#`.

```
$ pip install quokka-sharp
```

Given a quantum circuit in QASM form (`qasmfile1`), one can encode it as a weighted CNF file (`circ.cnf`) using the following python script.

```
import quokka_sharp as qk

circuit1 = qk.encoding.QASMparser(qasmfile1, True)
# Encode the circuit
cnf = qk.encoding.QASM2CNF(circuit1)
# Choose firstzero or allzero
cnf.add_measurement("firstzero")
# Export to benchmarks
cnf.write_to_file("circ.cnf")
```

Quantum circuit files can be downloaded from MQTBench by first choosing all possible quantum circuits and then following the instructions in Figure 1.

Since we only measure the first qubit in the current benchmarks, all measure command should be removed from the QASM files. For example:

```
measure q[0] -> meas[0];
measure q[1] -> meas[1];
```

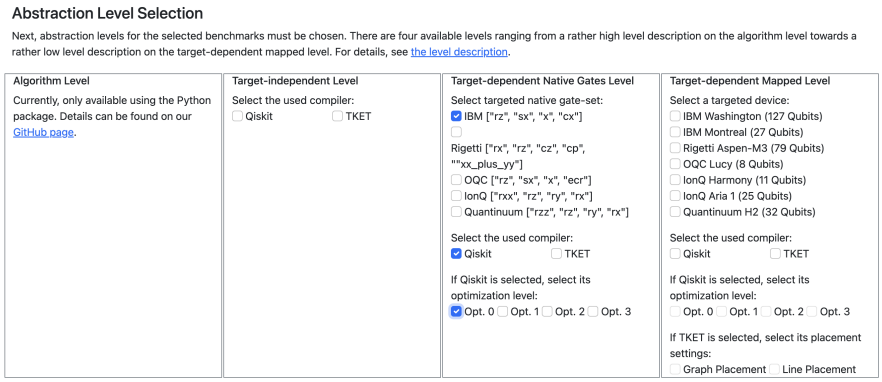


Figure 1: Choose target-dependent native gates-level and then use gate set IBM[“rz”,“sx”, “x”, “cx”], comipler Qiskit and set optimization level to be 0.

References

[1] Jingyi Mei, Marcello Bonsangue, and Alfons Laarman. Simulating quantum circuits by model counting, 2024.

[2] Nils Quetschlich, Lukas Burgholzer, and Robert Wille. Mqt bench: Benchmarking software and design automation tools for quantum computing. *Quantum*, 7:1062, July 2023.